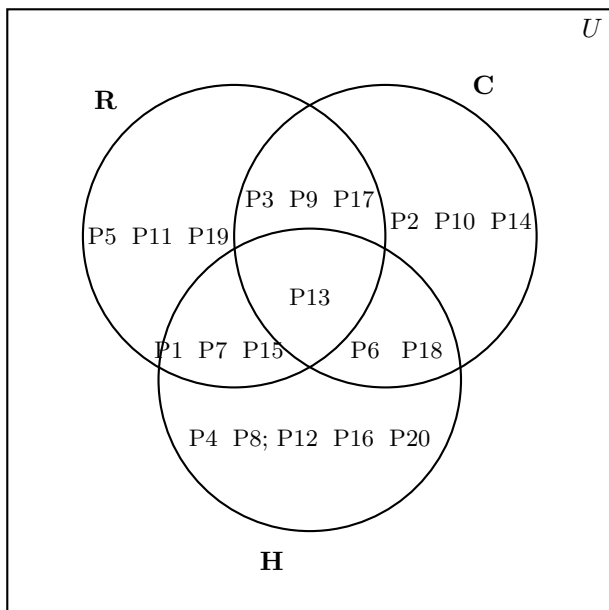




$$H \cap D \cap C = \{5, 6\}$$

$$H \cap D = \{3, 4, 5, 6\}$$

$$H \cap C = \{5, 6, 7, 8\}$$

$$D \cap C = \{5, 6, 12\}$$

$$\text{Solo } H = \{1, 2, 9, 10\}$$

$$\text{Solo } D = \{13, 14, 15\}$$

$$\text{Solo } C = \{16, 17, 19, 20\}$$

$$H \cup D \cup C =$$

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 19, 20\}$$

$$(H \cup D \cup C)^c = \{11, 18\}$$